

Slip No: 25

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination
Lab Course CS-308-MJ-P
Data Science and Analytics

Duration: 3 Hours

Maximum Marks:35

Q1. Consider the following dataset representing the number of students enrolled in different courses:
courses = ["BCA", "BBA", "B.Com", "B.Sc", "BA"]
students = [60, 45, 75, 50, 35]
Create a Bar Chart to visualize the number of students enrolled in each course. Use different bar colors, add an appropriate title, and label both axes.

[10 Marks]

Q2. Create two DataFrames containing information about different mobile phones.
DataFrame 1: Brand, Model, Price, RAM
DataFrame 2: Model, Storage, Battery_Capacity, Operating_System

Perform the following data transformation techniques:

- a) Data Integration by combining the two DataFrames using Model
 - b) Label Encoding on the Brand attribute of the merged DataFrame
 - c) One-Hot Encoding on the Operating_System attribute of the merged DataFrame
- Display the original DataFrames, merged DataFrame, and transformed DataFrame.

[20 Marks]

OR

Q2. Write a Python program to perform market basket analysis on the Groceries dataset using the Apriori algorithm. Read and explore the dataset, preprocess the data by handling missing values, convert categorical transaction data into a suitable numeric format, and generate frequent itemsets and association rules. Repeat the analysis using minimum support values 0.3 and 0.4.

[20 Marks]

S.No.	Items Purchased
1	Basmati Rice, Toor Dal, Turmeric Powder, Cooking Oil
2	Atta, Sugar, Tea, Milk
3	Rice, Moong Dal, Salt, Cooking Oil
4	Wheat Flour, Chickpeas, Turmeric Powder, Cumin Seeds
5	Poha, Peanuts, Onions, Cooking Oil
6	Basmati Rice, Paneer, Garam Masala, Tomatoes
7	Atta, Potatoes, Onions, Cooking Oil

Q3. Viva

[5 Marks]